

Harnal® OCAS 400mcg Film-coated Tablet (tamsulosin hydrochloride)

1. NAME OF THE MEDICINAL PRODUCT

Harnal OCAS, 400 micrograms, film-coated prolonged release tablet

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains as active ingredient tamsulosin hydrochloride 400 micrograms, equivalent to 367 micrograms tamsulosin.

For excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film coated, prolonged release tablet.

Round, yellow, film-coated tablets debossed with the code '04'.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of functional symptoms of benign prostatic hyperplasia (BPH).

4.2 Posology and method of administration

Posology

One tablet daily, to be taken with or without food.

Paediatric Population

The safety and efficacy of tamsulosin in children < 18 years have not been established. Currently available data are described in Section 5.1.

Method of administration

For oral use.

The tablet should be swallowed whole and should not be crunched or chewed as this will interfere with the prolonged release of the active ingredient.

4.3 Contraindications

A history of orthostatic hypotension; severe hepatic insufficiency.

Hypersensitivity to tamsulosin hydrochloride or any other component of the product.

4.4 Special warnings and precautions for use

As with other alpha1 blockers, a reduction in blood pressure can occur in individual cases during treatment with Harnal OCAS, as a result of which, rarely, syncope can occur. At the first signs of orthostatic hypotension (dizziness, weakness), the patient should sit or lie down until the symptoms have disappeared.

Before therapy with Harnal OCAS is initiated, the patient should be examined in order to exclude the presence of other conditions which can cause the same symptoms as benign prostatic hyperplasia. Digital rectal examination and, when necessary, determination of prostate specific antigen (PSA) should be performed before treatment and at regular intervals afterwards.

The treatment of severely renally impaired patients (creatinine clearance of less than 10 ml/min) should be approached with caution as these patients have not been studied.

The 'Intraoperative Floppy Iris Syndrome' (IFIS, a variant of small pupil syndrome) has been observed during cataract and glaucoma surgery in some patients on or previously treated with tamsulosin hydrochloride. IFIS may increase the risk of eye complications during and after the operation.

Discontinuing tamsulosin 1-2 weeks prior to cataract or glaucoma surgery is anecdotally considered helpful, but the benefit of treatment discontinuation has not been established. IFIS has also been reported in patients who had discontinued tamsulosin for a longer period prior to the surgery.

The initiation of therapy with tamsulosin hydrochloride in patients for whom cataract or glaucoma surgery is scheduled is not recommended. During pre-operative assessment, surgeons and ophthalmic teams should consider whether patients scheduled for cataract or glaucoma surgery are being or have been treated with tamsulosin in order to ensure that appropriate measures will be in place to manage the IFIS during surgery.

Tamsulosin hydrochloride should not be given in combination with strong inhibitors of CYP3A4 (e.g. ketoconazole) in patients with poor metaboliser CYP2D6 phenotype.

Tamsulosin hydrochloride should be used with caution in combination with strong (e.g. ketoconazole) and moderate (e.g. erythromycin) inhibitors of CYP3A4 (see section 4.5).

Cases of allergic reaction to tamsulosin in patients with a past history of sulfonamide allergy have been reported. If a patient reports a previously experienced sulfa allergy, caution is warranted when administering tamsulosin hydrochloride.

4.5 Interaction with other medicinal products and other forms of interaction

No interactions have been seen when tamsulosin was given concomitantly with atenolol, enalapril or theophylline. Concomitant cimetidine brings about a rise in plasma levels of tamsulosin, and furosemide a fall, but as levels remain within the normal range, posology need not be changed.

In vitro, neither diazepam nor propranolol, trichlormethiazide, chlormadinon, amitriptyline, diclofenac, glibenclamide, simvastatin and warfarin change the free fraction of tamsulosin in human plasma. Neither does tamsulosin change the free fractions of diazepam, propranolol, trichlormethiazide, and chlormadinon.

Diclofenac and warfarin, however, may increase the elimination rate of tamsulosin.

Concomitant administration of tamsulosin hydrochloride with strong inhibitors of CYP3A4 may lead to increased exposure to tamsulosin hydrochloride. Concomitant administration with ketoconazole (a known strong CYP3A4 inhibitor) resulted in an increase in AUC and Cmax of tamsulosin hydrochloride by a factor of 2.8 and 2.2, respectively.

Tamsulosin hydrochloride should be used with caution in combination with strong (e.g. ketoconazole) and moderate inhibitors (e.g. erythromycin) of CYP3A4.

Concomitant administration of tamsulosin hydrochloride with paroxetine, a strong inhibitor of CYP2D6, resulted in a Cmax and AUC of tamsulosin that had increased by a factor of 1.3 and 1.6, respectively, but these increases are not considered clinically relevant.

There is a theoretical risk of enhanced hypotensive effect when given concurrently with drugs which may reduce blood pressure, including anaesthetic agents and other α_1 -adrenoceptor antagonists.

4.6 Pregnancy and lactation

Harnal OCAS is not indicated for use in women.

Ejaculation disorders have been observed in short and long term clinical studies with tamsulosin. Events of ejaculation disorder, retrograde ejaculation and ejaculation failure have been reported in the post authorization phase.

4.7 Effects on ability to drive and use machines

No data is available on whether Harnal OCAS adversely affects the ability to drive or operate machines. However, in this respect patients should be aware of the fact that drowsiness, blurred vision, dizziness and syncope can occur.

4.8 Undesirable effects

System Organ Class	Common >1/100, <1/10	Uncommon >1/1000, <1/100	Rare >1/10,000, <1/1000	Very Rare <1/10,000	Not known (cannot be estimated from the available data)
Nervous system disorders	dizziness (1.3%)	headache	syncope		
Eye disorders					Vision blurred* Visual impairment*
Cardiac disorders		palpitations			
Vascular disorders		orthostatic hypotension			
Respiratory, thoracic and mediastinal disorders	Nasal congestion	rhinitis			Epistaxis*
Gastrointestinal disorders		constipation, diarrhea, nausea, vomiting			Dry mouth*
Skin and subcutaneous tissue disorders		rash, pruritus, urticaria	angioedema	Stevens-Johnson syndrome	Erythema multiforme* Dermatitis exfoliative*
Reproductive system and breast disorders	ejaculation disorders including retrograde ejaculation and ejaculation failure			priapism	
General disorders and administration site conditions		asthenia			

*observed post-marketing

As with other alpha-blockers, drowsiness, blurred vision, dry mouth or oedema can occur.

During cataract and glaucoma surgery a small pupil situation, known as Intraoperative Floppy Iris Syndrome (IFIS), has been associated with therapy of tamsulosin during post-marketing surveillance (see also section 4.4).

Post-marketing experience: In addition to the adverse events listed above, atrial fibrillation, arrhythmia, tachycardia and dyspnoea have been reported in association with tamsulosin use. Because these spontaneously reported events are from the worldwide post marketing experience, the frequency of events and the role of tamsulosin in their causation cannot be reliably determined.

4.9 Overdose

Symptoms

Overdosage with tamsulosin hydrochloride can potentially result in severe hypotensive effects, dizziness and malaise. Severe hypotensive effects have been observed at different levels of overdosing.

Treatment

In case of acute hypotension occurring after overdose, cardiovascular support should be given. Blood

pressure can be restored and heart rate brought back to normal by lying the patient down. If this does not help, then volume expanders, and when necessary, vasopressors could be employed. Renal function should be monitored and general supportive measures applied. Dialysis is unlikely to be of help, as tamsulosin is very highly bound to plasma proteins.

Measures, such as emesis, can be taken to impede absorption. When large quantities are involved, gastric lavage can be applied and activated charcoal and an osmotic laxative, such as sodium sulphate, can be administered.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group:

Alpha1-adrenoceptor antagonist.

ATC code: G04C A02. Preparations for the exclusive treatment of prostatic disease.

Mechanism of action:

Tamsulosin binds selectively and competitively to postsynaptic alpha1-receptors, in particular to the subtype alpha1A, which bring about relaxation of the smooth muscle of the prostate, whereby tension is reduced.

Pharmacodynamic effects:

Harnal OCAS increases maximum urinary flow rate by reducing smooth muscle tension in the prostate and urethra, thereby relieving obstruction.

It also improves the complex of irritative and obstructive symptoms in which bladder instability and tension of the smooth muscles of the lower urinary tract play an important role. Alpha1-blockers can reduce blood pressure by lowering peripheral resistance. No reduction in blood pressure of any clinical significance was observed during studies with Harnal OCAS.

Paediatric Population

A double-blind, randomized, placebo-controlled, dose ranging study was performed in children with neuropathic bladder. A total of 161 children (with an age of 2 to 16 years) were randomized and treated at 1 of 3 dose levels of tamsulosin (low [0.001 to 0.002 mg/kg], medium [0.002 to 0.004 mg/kg], and high [0.004 to 0.008 mg/kg]), or placebo. The primary endpoint was number of patients who decreased their detrusor leak point pressure (LPP) to <40 cm H₂O based upon two evaluations on the same day. Secondary endpoints were: Actual and percent change from baseline in detrusor leak point pressure, improvement or stabilization of hydronephrosis and hydroureter and change in urine volumes obtained by catheterisation and number of times wet at time of catheterisation as recorded in catheterisation diaries. No statistically significant difference was found between the placebo group and any of the 3 tamsulosin dose groups for either the primary or any secondary endpoints. No dose response was observed for any dose level.

5.2 Pharmacokinetic properties

Absorption:

Harnal OCAS is formulated as an Oral Controlled Absorption System (OCAS) and is a prolonged release tablet of the non-ionic gel matrix type.

Tamsulosin administered as prolonged release tablets is absorbed from the intestine. Under fasting conditions approximately 57% of the administered dose is estimated to be absorbed. A consistent slow release of tamsulosin is maintained over the whole pH range encountered in the gastro intestinal tract with little fluctuation over 24 hours. The extent of absorption is increased by 64% and 149% (AUC and C_{max} respectively) by a high fat meal compared to fasted.

Tamsulosin shows linear kinetics.

After a single dose of Harnal in the fasted state, plasma levels of tamsulosin peak at a median time of 6 hours. In steady state, which is reached by day 4 of multiple dosing, plasma levels of tamsulosin peak at 4 to 6 hours in the fasted and fed state. Peak plasma levels increase from approximately 6

ng/ml after the first dose to 11 ng/ml in steady state.

As a result of the prolonged release characteristics of Harnal OCAS, the trough concentration of tamsulosin in plasma amounts to 40% of the peak plasma concentration under fasted and fed conditions.

There is a considerable inter-patient variation in plasma levels, both after single and multiple dosing.

Distribution:

In man, tamsulosin is about 99% bound to plasma proteins and volume of distribution is small (about 0.2l/kg).

Metabolism:

Tamsulosin has a low first pass effect, being metabolised slowly. Most tamsulosin is present in plasma in the form of unchanged drug. It is metabolised in the liver.

In rats, hardly any induction of microsomal liver enzymes was seen to be caused by tamsulosin.

In vitro results suggest that CYP3A4 and also CYP2D6 are involved in metabolism, with possible minor contributions to tamsulosin hydrochloride metabolism by other CYP isozymes. Inhibition of CYP3A4 and CYP2D6 drug metabolizing enzymes may lead to increased exposure to tamsulosin hydrochloride (see Section 4.4 and 4.5).

No dose adjustment is warranted in hepatic insufficiency.

None of the metabolites are more active than the original compound.

Elimination:

Tamsulosin and its metabolites are mainly excreted in the urine. The amount excreted as unchanged drug is estimated to be about 4 - 6% of the dose, administered as Harnal OCAS.

After a single dose of Harnal OCAS, and in steady state, elimination half-lives of about 19 and 15 hours, respectively, have been measured.

No dose adjustment is necessary in patients with renal impairment.

5.3 Preclinical safety data

Single and repeat dose toxicity studies were performed in mice, rats and dogs. In addition, reproduction toxicity studies were performed in rats, carcinogenicity in mice and rats, and in vivo and in vitro genotoxicity were examined. The general toxicity profile, as seen with high doses of tamsulosin, is consistent with the known pharmacological actions of the alpha-adrenergic blocking agents. At very high dose levels, the ECG was altered in dogs. This response is considered to be not clinically relevant. Tamsulosin showed no relevant genotoxic properties.

Increased incidences of proliferative changes of mammary glands of female rats and mice have been reported. These findings, which are probably mediated by hyperprolactinaemia and only occurred at high dose levels, are regarded as irrelevant.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Core

Macrogol (containing butylhydroxytoluene)
Magnesium stearate

Film-coat

Hypromellose
Macrogol
Yellow iron oxide (E172)

6.2 Incompatibilities

None known.

6.3 Shelf life

The shelf life is indicated on the packaging.

6.4 Special condition

Store at or below 30°C.

6.5 Nature and contents of container

Aluminium foil blister packs containing either 10 tablets or 30 tablets.

6.6 Special precautions for disposal and other handling

No special instructions.

7. MANUFACTURER

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The Netherlands

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